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1. POWERPLANT BACKGROUND AND CONTEXT

1.1 Summarized Engine Information

The latest addition to the CF34-engine family is the General Electric (GE) CF34-8 Turbofan Engine, which powers the Bombardier CRJ-700 aircraft. The engine has a thrust capacity of 14500 lbs and can effectively power the 90-passenger jet.

To accommodate the larger and longer-range regional jets, the GE CF34-8 engine (CRJ-700/900 series) includes newer innovations over the former engines, including: 'a current split fan for enhanced bypass ratio and thrust; a 10-stage high-efficiency high-pressure compressor with an overall pressure ratio (pr) of 28:1; and a low-emission combustor for environmental compliance' (USITC, 2004).

The addition of a digital control to more precisely manage the system as well as the use of materials more resistant to thermal shocks and wear allows for higher operational temperatures. A few problems that arose with then implementation of this engine to the CRJ-700 include, but are not limited to : added mass due to a the engine being heavier than its predecessor, the CF34-3, excessive thermal stresses due to elevated temperatures. Overall, the CF34-8 integrated reliable, efficient, and environmentally compliant engines for the new regional aircraft. [1]

A table that summarizes the augmentations, drawbacks and benefits and measures to compensate for the engine augmentations is presented below

Table 1: Summary of all augmentations

Augmentation	Benefit	Drawback	Compensation
Larger fan	Increased thrust and engine efficiency	Increased mass and drag	Lighter blades and a stronger nacelle
Higher Pressure Ratio (Overall)	Better thermal efficiency	Increased Thermal Stress	Cooling, through variable stator vanes
Turbine cooling & materials	Aircraft can generate more thrust. Improving thrust-to-weight ratio.	Increased Maintenance costs	Materials with higher resistance to thermal shocks and stresses



Figure 1: CF34-8 Powerplant. [1]

1.2 Design Selection

1.2.1 Specific Fuel Consumption and Cost

The CRJ-700 is designed for medium to long haul flights, based on the mission profile of the aircraft the engine would be ideal in that it produces large amounts of thrust with a low specific fuel consumption. Given that the aviation industry is highly competitive, the aircraft that can fly the furthest with the least amount of fuel will always be preferable and that simply stems from the fact that flying such an aircraft is far less expensive for the airline and the customer, which in effect, will garner more customers to the airline operating the aircraft.

Proof of this claim is present below in the form of an SFC chart that was developed for the CRJ-700 along with two of its other competitors. The chart highlights the fact that the CRJ-700 consumes less fuel per kilogram of thrust per hour of flight.

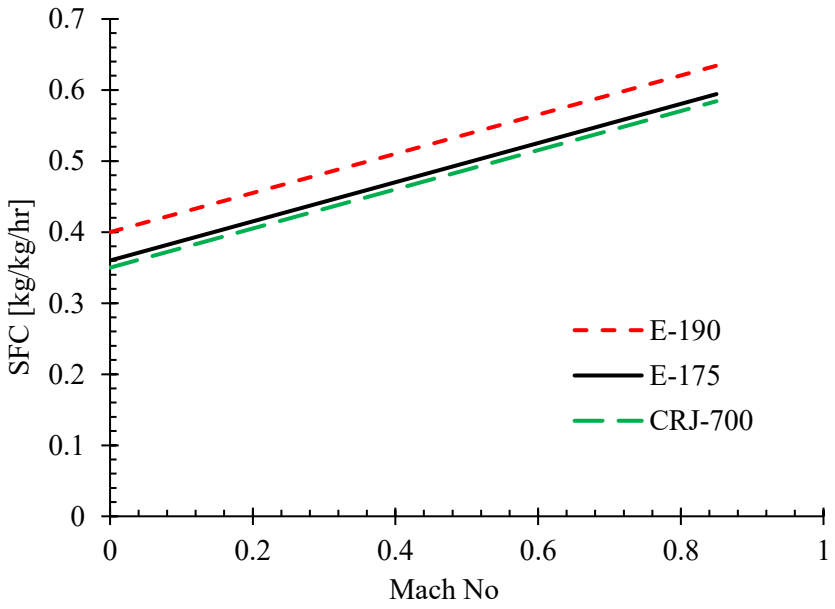


Figure 2: SFC of curves for CRJ-700 vs competition

Based on this information, it can be argued that, that is the reason why Bombardier chosen the CF34-8 as its powerplant.

1.2.2 Airframe and Thrust to Weight

The CRJ-700 is a modified CRJ-200 which is elongated in order to increase the carrying capacity of the aircraft.

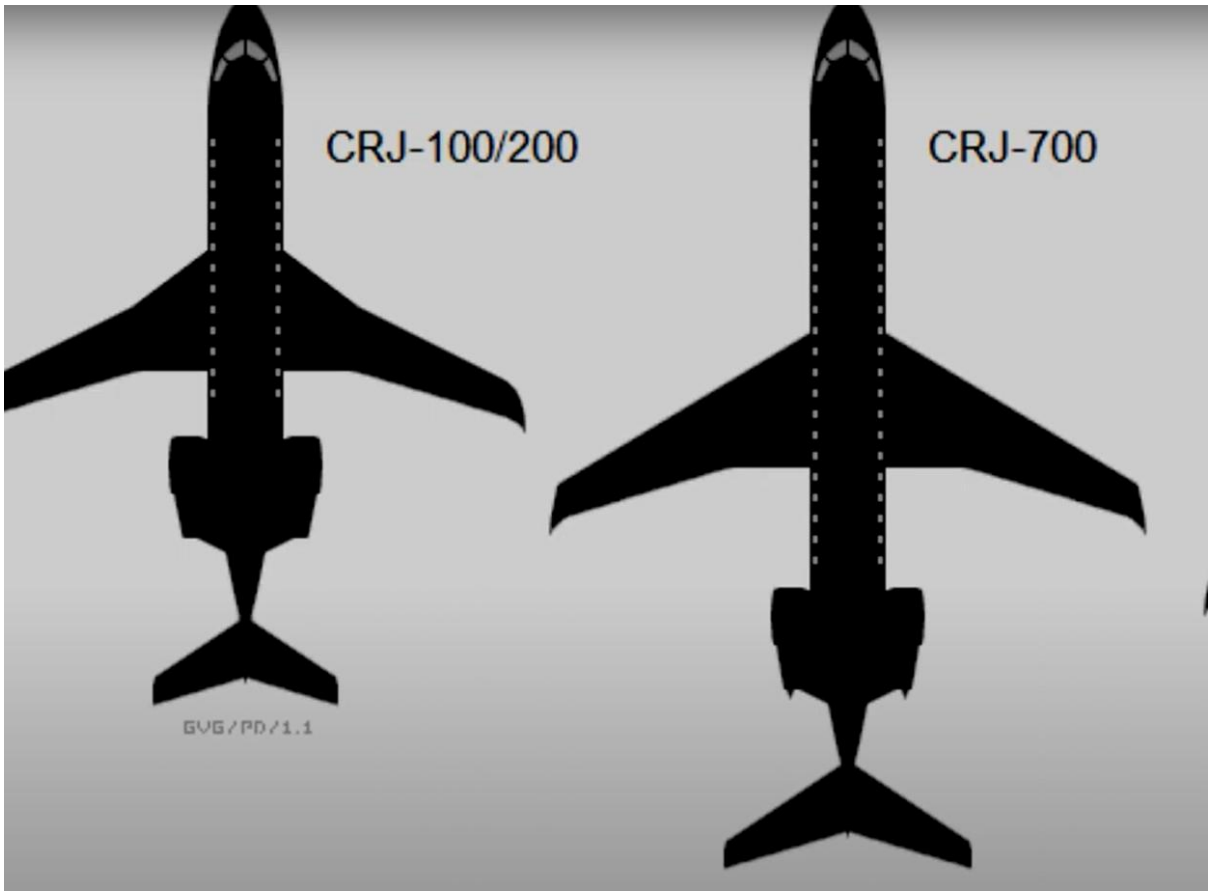


Figure 3: CRJ-200 vs CRJ-700. [2]

The CF34-8 shares a core architecture with the CF34-3 (used on the CRJ-200) but with improved thrust and performance. Structurally, this meant minimal changes to the nacelle, pylon, and systems while gaining higher performance. The CRJ-700, like earlier CRJs, uses rear-mounted engines.

The size of the CF34-8 and mounting configuration are ideal for pylon mounting on the rear fuselage without requiring structural redesign of the tail or fuselage frame. In addition the produced thrust of 14500 lbs balances well with the mass of the aircraft allowing for efficient takeoff without excessive stress to the airframe. [1]

2. ASSUMPTIONS

- Changes in potential and kinetic energy are negligible.
- The pressure at the diffuser inlet and nozzle exit are equal to the ambient pressure at any given altitude (i.e. $P_1 = P_6 = P_{atm}$).
- The engine has 10 High Pressure compressor stages, 4 Low Pressure and 2 High Pressure turbine stages. For ease of analysis, all the stages for the compressor were amalgamated into a single compressor. The same approach was applied to the Turbine.

- All thermodynamic processes occurring within the powerplant were assumed to be under steady state conditions, no transient effects were included in the analysis.
- The velocity close to the compressor inlet was approximated to be 0 m/s.
- The turbine inlet was assumed to operate at a temperature of 1130 K at sea-level and 980 K at cruise. [3]

- The nozzle and diffuser were assumed to operate under isentropic conditions. Whilst the compressor and turbine were assumed to operate under isobaric conditions.
- The engine does not have an intercooler, preheater and an afterburner. Therefore reheat and intercooling effects were neglected from the analysis.
- Static sea-level calculations were performed using published maximum rated thrust to help simulate take-off conditions as best as possible.

- Working Fluid is air. Which, in addition, is assumed to be an ideal gas.
- Constant fan performance, free of frictional and added heat effects.
- Constant air density at any given altitude.

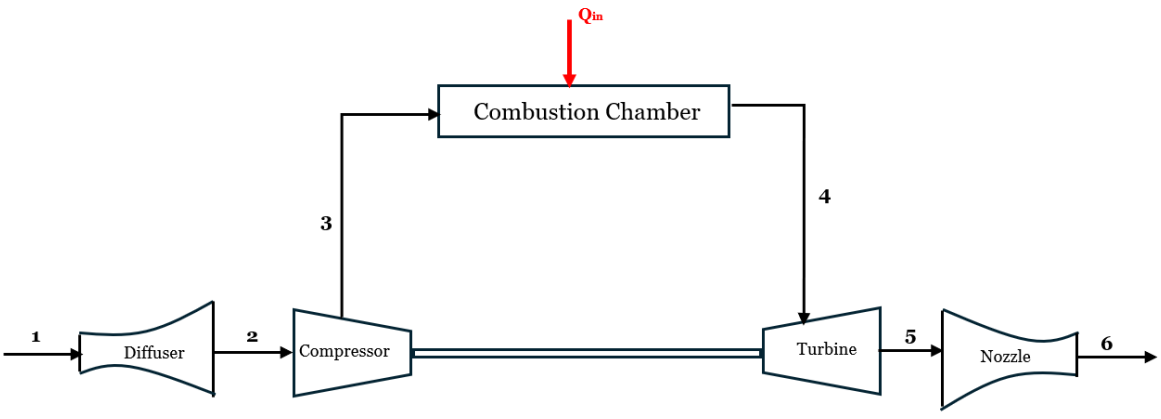


Figure 4: Open Air-Brayton Cycle for CF34-8 Engine.

3. METHOLOGY

3.1 Sea Level

The analysis was done in an iterative manner, with the assumption that the compressor and turbines were 95 % and 90 % efficient, as a first estimate. The inlet velocity, V_1 was assumed to be 0 m/s, as the aircraft was assumed to be static at sea-level. In addition, the ambient pressure and temperatures were taken as 101.235 KPa and 300 K at Sea-level. The thermodynamic table A-22 along with the energy balance equation (that relates enthalpy to speed) were used to estimate the pressure ratios, enthalpy values and temperatures for each station as outlined in the schematic. A constant mass flow of 300 kg/s, at sea-level, was assumed initially. The exit speed and the mass flow were used to estimate the thrust.

After several iterations the following table was developed.

Table 2: Engine Thermodynamic Analysis at Sea Level.

Station	V [m/s]	P [KPa]	T [K]	PR	H [kg/kJ]	Hs [kg/kJ]
1	0.00	101.33	300.00	1.39	300.19	0.00
2	-	101.33	300.00	1.39	300.19	0.00
3	-	405.30	480.56	7.30	483.06	446.49
3s	-	2877.63	897.14	74.39	929.73	840.40
4	-	2877.63	1130.00	186.40	1195.93	0.00
5	-	357.01	733.36	34.32	749.26	670.43
5s	-	107.98	561.17	12.76	566.39	534.11
6	289.52	101.33	520.82	9.74	524.48	0.00

The exit thrust was estimated to be 66589 N, when concerted to pounds is equal to 14969.80 lbs, with an exit velocity V_6 of 289.52 m/s. The mass final mass flow rate sea-level was estimated to be 240 kg/s.

The compressor and turbine efficiency values that yielded the best results post analysis were found to be 80% and 85% respectively. These values were applied at both sea-level and at cruise. T-s diagram at sea-level conditions inclusive of irreversibility effects is presented below.

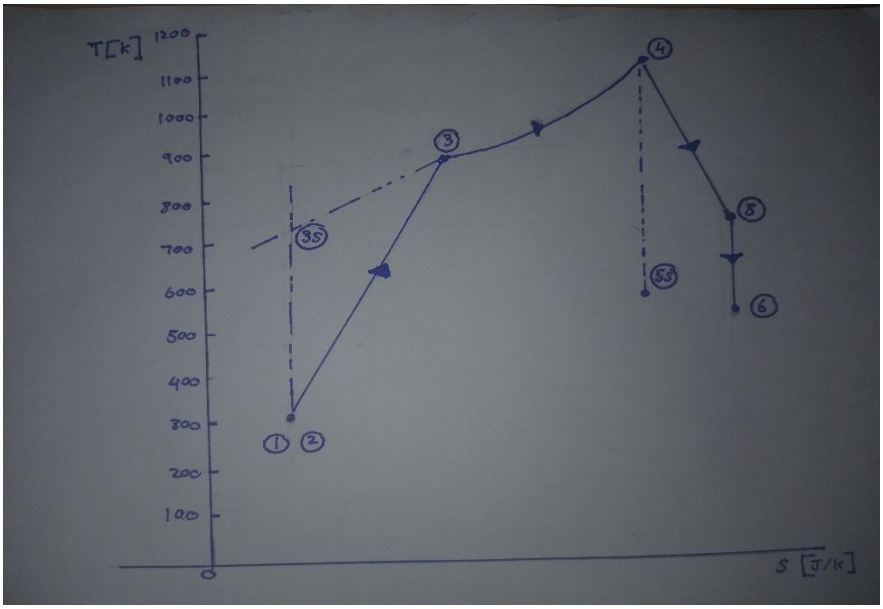


Figure 5: Sea-Level T-s Diagram.

3.2 Cruise Condition

The above analysis was done at cruise. From literature the CRJ-700 cruises at 36000 ft at an approximate speed of 243 m/s [4]. Ambient temperature and pressure values at this altitude were extracted using US-standard temperature and pressure table, in Appendix A. After which, a data table was generated in the same manner as before and is presented below:

Station	V [m/s]	P [Kpa]	T [K]	PR	H [kg/kJ]	Hs [kg/KJ]
1	252.00	23.84	216.83	0.45	216.80	0.00
2	-	38.34	248.51	0.72	248.55	0.00
3s	-	153.35	399.35	3.78	400.32	369.96
3	-	1088.76	756.11	38.52	773.94	699.22
4	-	1088.76	980.00	105.20	1023.25	0.00
5	-	146.78	640.38	20.69	649.62	583.69
5s	-	47.43	494.98	8.12	497.86	471.07
6	469.77	23.84	386.69	3.36	387.51	0.00

The exit thrust was estimated to be 46316 N, when concerted to pounds is equal to 10412.25 lbs, with an exit velocity V_6 of 469.77 m/s. The associated T-s diagram at sea-level conditions inclusive of irreversibility effects is presented below.

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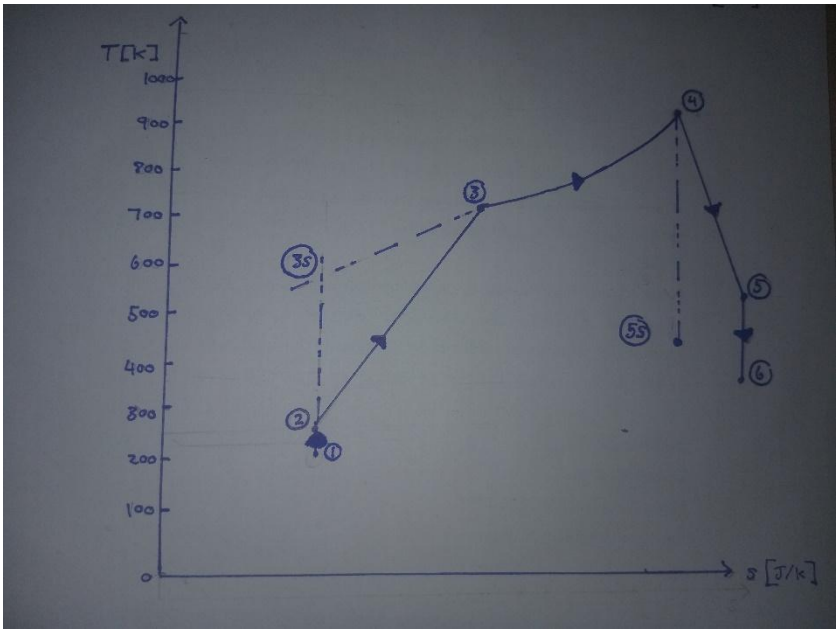


Figure 6: Cruise T-s diagram.

4 CRITICAL DISCUSSION OF RESULTS

4.1 Sea-Level

Under sea-level conditions, the engine operates at maximum thrust and maximum fuel flow, aiming to dominate thrust output whilst overlooking fuel economy. This mode is especially important during takeoff when significant inertia of the CRJ has to be overcome, and enough lift must be generated relative to aircraft mass over a fairly short runway.

Input parameters in the analysis consisted of a static takeoff with an inlet velocity of 0 m/s, an ambient temperature of 288.15 K, and standard atmospheric pressure of 101.325 kPa. The dense atmospheric conditions and full throttle operation of the engine gives the expected thrust output at sea level which, as anticipated, greatly exceeds the value at cruise altitude.

With the engine's fuel mass flow rate at this point of the flight envelope, the combustion chamber temperature to assumed to be 1130 K, in addition a fuel mass flow rate of around 1.03 kg/s was assumed to sustain the energy input at the combustor, which is required for the engine to be able to produce expansive jet acceleration at the nozzle, while increasing the enthalpy drop across the turbines as the working fluid is expelled, for optimal thrust. To further maximize thrust, the air-mass flow rate was increased to an assumed 300 kg/s during the iteration process. This would allow the engine to take advantage of the denser air at sea-level thus simulating realistic takeoff conditions.

4.2 Cruise Discussion

At cruising conditions, the power of the engine represents a compromise between efficiency and thrust retention. The cruise inputs were that the inlet velocity would be 240 m/s Mach (0.78) where the ambient temperature is an estimated 216.65 K, along with an ambient pressure of 23.84 KPa. The resultant thrust output was significantly less than the sea-level take-off thrust. Which is consistent with the engine behaviour at such a high altitude [4].

That is because at cruise, the engine is throttled back to lower consumption in order to increase the range and endurance of the aircraft. Which is necessary based in the fact that the specific fuel consumption increases linearly with the Mach number, for the CRJ-700 at Mach 0.78, according to figure 2, the specific fuel consumption would be about 0.5 kg/kg/hr.

To combat the high altitude, lower density and temperature, the engine sustains an approximate turbine inlet temperature of 980 K, in order to guarantee proper expansion through the turbines and reasonably high propulsion. The overall pressure ratio (OPR) still remains high at approximately 28.40:1 because the compressors were assumed to continuously compress the air at their estimated peak efficiency of 80% without any lag or disturbances.

4 COMPARSION OF DATA AND CRITICAL ANALYSIS

4.1 Summary Results

In addition to the table method applied. Thrust performance of the CRJ-700 was estimated using the Jenkinson Thrust curves, in appendix A, for a BPR (Bypass Ratio) of 5, at sea-level and at various altitudes. Although this is out of scope for the assignment, the benefit of this analysis lies in the fact that it helped increased the robustness of the investigation by adding another element that could help validate the published results.

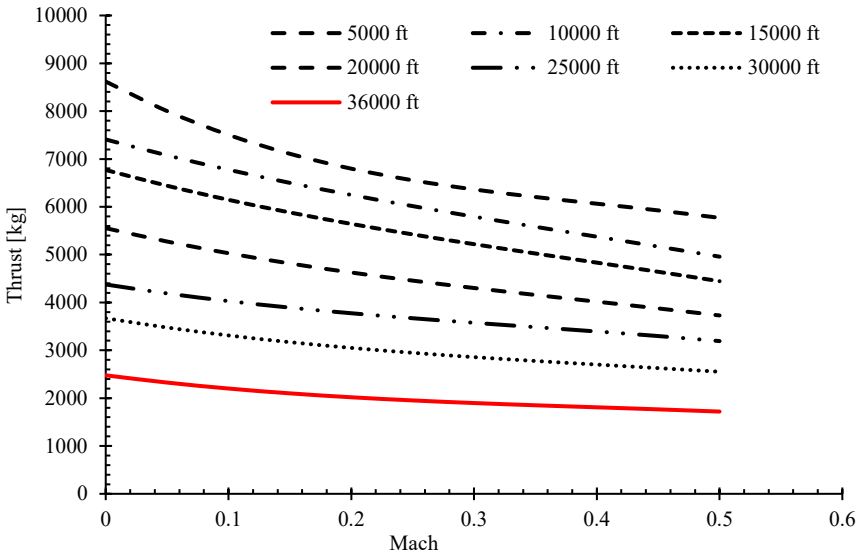


Figure 7:Thrust curves at various altitudes.

The thrust at the altitude of interest, at 36000 ft is highlighted in red. The maximum thrust of 2478 kg, which is 5463.055 lbs.

The Sea-level thrust curve developed is as follows:

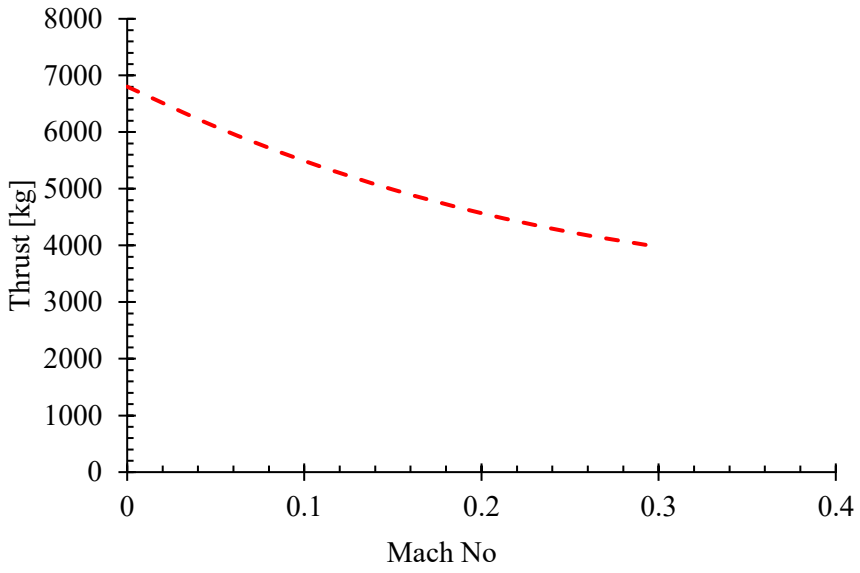


Figure 8: Sea-Level Thrust Curve.

The maximum developed thrust as per the Jenkinson estimate yielded a thrust estimate of 6083.224 kg which is approximately 14998.54 lbs.

4.2 Comparison of Results

Table 3: Comparative Table [1]

	Thrust - Sea Level [lbs]	Thrust - Cruise (Max) [lbs]	Pressure Ratio - Overall	Bypass Ratio (BPR)
Published	14500.00	-	28.00	5
Estimated (Hand Calc)	14969.80	10412.25	28.40	7.1
Estimated (Jenkinson)	14998.54	5463.06	-	-

4.2.1 Thrust comparison

The error between the published and estimated metrics was quite minimal. Indicating that the efficiency estimates and the approaches used were fairly accurate. With there being an approximate 3.3% margin of error for thrust estimates at sea level.

The margin of error between the Jenkinson estimate and the hand calculation was an approximate 47%, the Jenkinson estimate was assumed to be the less accurate of the two at cruise because it does not take into account mass flow, pressure ratios and other thermodynamic variations. And because no published cruise thrust data could be obtained for the thrust at cruise. The hand calculated value was assumed to be the best thrust estimate. Based in the fact that there was good correlation between the sea-level thrust and the published thrust.

4.2.2 Overall Pressure Ratio (OPR) Comparison

The overall pressure ratio was estimated by dividing the pressure of the working fluid (air) just before it enters the combustion chamber inlet, with the inlet nozzle pressure. Which yielded a 1% error margin between the published and estimated results.

4.2.3 Bypass ratio comparison

The bypass ratio was estimated by diving the pressure at station 4 with the pressure at station 3. The error between the estimated bypass ratio the published bypass ratio was estimated to be 30%, which is excessive to say the least. The source of this error was determined to most likely stem from assumed mass flowrate of 240 kg/s at cruise during the iterative process. Another major assumption that most likely contributed to the error, is that constant air density and fan performance were assumed, theses assumptions account for real-world losses or flow restrictions in the bypass duct. These omissions prevent real-world effects from being captured, hence there was bound to be some form of error.

APPENDIX A

Table 4: Elevation vs Thermodynamic Properties (US-standard) [4].

Geo potential Altitude above Sea Level ***- h -*** ***(m)***	**Temperature** ***- t -*** ***(°C)***	Acceleration of Gravity ***- g -*** ***(m/s^2)***	**Absolute Pressure** ***- p -*** ***(10^4 N/m^2)***	**Density** ***- ρ -*** ***(kg/m^3)***	Dynamic Viscosity ***- μ -*** ***(10^{-5} $N s/m^2$)***
-1000	21.5	9.81	11.39	1.347	1.821
0	15	9.807	10.13	1.225	1.789
1000	8.5	9.804	8.988	1.112	1.758
2000	2	9.801	7.95	1.007	1.726
3000	-4.49	9.797	7.012	0.9093	1.694
4000	-10.98	9.794	6.166	0.8194	1.661
5000	-17.47	9.791	5.405	0.7364	1.628
6000	-23.96	9.788	4.722	0.6601	1.595
7000	-30.45	9.785	4.111	0.59	1.561
8000	-36.94	9.782	3.565	0.5258	1.527
9000	-43.42	9.779	3.08	0.4671	1.493
10000	-49.9	9.776	2.65	0.4135	1.458
15000	-56.5	9.761	1.211	0.1948	1.422
20000	-56.5	9.745	0.5529	0.08891	1.422
25000	-51.6	9.73	0.2549	0.04008	1.448
30000	-46.64	9.715	0.1197	0.01841	1.475
40000	-22.8	9.684	0.0287	0.003996	1.601
50000	-2.5	9.654	0.007978	0.001027	1.704
60000	-26.13	9.624	0.002196	0.0003097	1.584
70000	-53.57	9.594	0.00052	0.00008283	1.438
80000	-74.51	9.564	0.00011	0.00001846	1.321

TABLE A-22 Ideal Gas Properties of Air

T(K), h and u(kJ/kg), s° (kJ/kg · K)												
				when $\Delta s = 0$[1]							when $\Delta s = 0$	
T	h	u	s°	p_r	v_r	T	h	u	s°	p_r	v_r
200	199.97	142.56	1.29559	0.3363	1707.	450	451.80	322.62	2.11161	5.775	223.6
210	209.97	149.69	1.34444	0.3987	1512.	460	462.02	329.97	2.13407	6.245	211.4
220	219.97	156.82	1.39105	0.4690	1346.	470	472.24	337.32	2.15604	6.742	200.1
230	230.02	164.00	1.43557	0.5477	1205.	480	482.49	344.70	2.17760	7.268	189.5
240	240.02	171.13	1.47824	0.6355	1084.	490	492.74	352.08	2.19876	7.824	179.7
250	250.05	178.28	1.51917	0.7329	979.	500	503.02	359.49	2.21952	8.411	170.6
260	260.09	185.45	1.55848	0.8405	887.8	510	513.32	366.92	2.23993	9.031	162.1
270	270.11	192.60	1.59634	0.9590	808.0	520	523.63	374.36	2.25997	9.684	154.1
280	280.13	199.75	1.63279	1.0889	738.0	530	533.98	381.84	2.27967	10.37	146.7
285	285.14	203.33	1.65055	1.1584	706.1	540	544.35	389.34	2.29906	11.10	139.7
290	290.16	206.91	1.66802	1.2311	676.1	550	554.74	396.86	2.31809	11.86	133.1
295	295.17	210.49	1.68515	1.3068	647.9	560	565.17	404.42	2.33685	12.66	127.0
300	300.19	214.07	1.70203	1.3860	621.2	570	575.59	411.97	2.35531	13.50	121.2
305	305.22	217.67	1.71865	1.4686	596.0	580	586.04	419.55	2.37348	14.38	115.7
310	310.24	221.25	1.73498	1.5546	572.3	590	596.52	427.15	2.39140	15.31	110.6
315	315.27	224.85	1.75106	1.6442	549.8	600	607.02	434.78	2.40902	16.28	105.8
320	320.29	228.42	1.76690	1.7375	528.6	610	617.53	442.42	2.42644	17.30	101.2
325	325.31	232.02	1.78249	1.8345	508.4	620	628.07	450.09	2.44356	18.36	96.92
330	330.34	235.61	1.79783	1.9352	489.4	630	638.63	457.78	2.46048	19.84	92.84
340	340.42	242.82	1.82790	2.149	454.1	640	649.22	465.50	2.47716	20.64	88.99
350	350.49	250.02	1.85708	2.379	422.2	650	659.84	473.25	2.49364	21.86	85.34
360	360.58	257.24	1.88543	2.626	393.4	660	670.47	481.01	2.50985	23.13	81.89
370	370.67	264.46	1.91313	2.892	367.2	670	681.14	488.81	2.52589	24.46	78.61
380	380.77	271.69	1.94001	3.176	343.4	680	691.82	496.62	2.54175	25.85	75.50
390	390.88	278.93	1.96633	3.481	321.5	690	702.52	504.45	2.55731	27.29	72.56
400	400.98	286.16	1.99194	3.806	301.6	700	713.27	512.33	2.57277	28.80	69.76
410	411.12	293.43	2.01699	4.153	283.3	710	724.04	520.23	2.58810	30.38	67.07
420	421.26	300.69	2.04142	4.522	266.6	720	734.82	528.14	2.60319	32.02	64.53
430	431.43	307.99	2.06533	4.915	251.1	730	745.62	536.07	2.61803	33.72	62.13
440	441.61	315.30	2.08870	5.332	236.8	740	756.44	544.02	2.63280	35.50	59.82

References

[1] G. Electric, “GE Aviation,” General Electric, 2024.

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